

UNIT : 04

Test of Hypothesis (Accept / reject)

Statistical Hypothesis: Assumption about any statistical data.

Hypothesis: 10% student of class are overweight.

n                      overweight

100                      8 %

150                      8.5 %

250                      7.2 %

300                      13 %

Result not matched and hence

if sampling error then accepted else rejected.

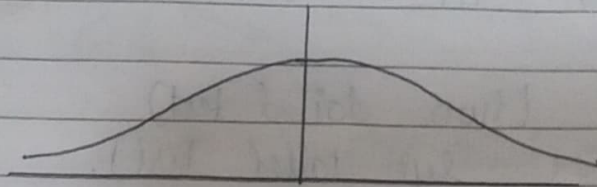
# Statistical Hypothesis:→ Population: aggregation of all objects, people, items, students under study.→ Sample: small portion of population or subset of population.→ Parameter: Any statistical measure obtained from population.  $\mu$ ,  $N$ ,  $\sigma$ ,  $\rho$  these are some population parameters.→ Statistic: Any statistical measure obtained from sample.  $\bar{X}$ ,  $n$ ,  $S$  (s.d of sample),  $r$  (correlation of sample), these are some sample parameters.→ Sampling distribution: distribution of all statistics of all samples. (frequency distribution).

→ standard error <sup>(SE)</sup>: standard deviation of sampling distribution.

$$SE(\bar{x}) = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$$

→ In large sample the distribution is distributed normally.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{\bar{x} - E(x)}{SE(x)}$$

→ level of significance <sup>(LOS)</sup>: maximum probability of rejecting the null hypothesis.

→ Hypothesis: Any claim, assumption or assertion about the population parameter or statistic.

There are two types of hypothesis:

1. Null hypothesis:  $H_0$  → some population parameter.

$$H_0: \theta_0 = \theta$$

$\theta_0$  → sample statistic.

Null hypothesis says that claim is true.  $H_0: \theta_0 = \theta$ , there is no significant difference between population parameter and sample statistic and difference if any is due to fluctuation of sampling.

$n > 30$  then Z test (large sample theory)  
and for less than 30 T test is applied (no normal distribution)  
Date: / / Page no:

2. Alternative Hypothesis :

$H_1 = \theta_0 \neq \theta$  significant difference between  
population parameter and statistical sampling  
 $\theta_0 < \theta$   
 $\theta_0 > \theta$

$H_1$  is just opposite of  $H_0$  to prove  
figure wrong

Example  $H_0 : \theta_0 \leq 10\%$   
then  $H_1 : \theta_0 > 10\%$

- $\theta_0 \neq \theta$  (two tailed test)
- $\theta_0 < \theta$  (left tailed test).
- $\theta_0 > \theta$  (right tailed test).

Almost 25% students are not placed.

$H_0 : \theta_0 \leq \theta = 25\%$

$H_1 : \theta_0 > \theta = 25\%$  (right tailed  
test).  
to prove hypothesis wrong

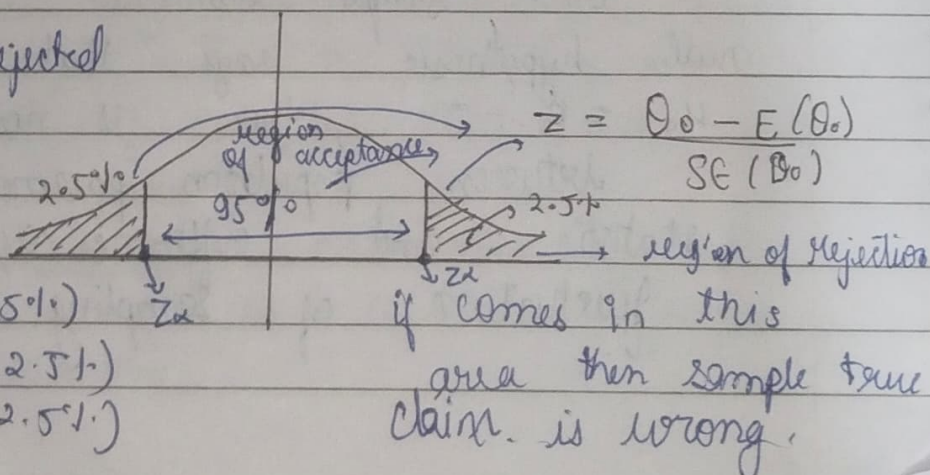
→ level of significance ( $\alpha$ ) : Maximum probability  
of rejecting a null hypothesis.  
It is denoted by  $\alpha$ .

st.  $H_0$  accepted  $H_1$  rejected

vice versa.

critical value.

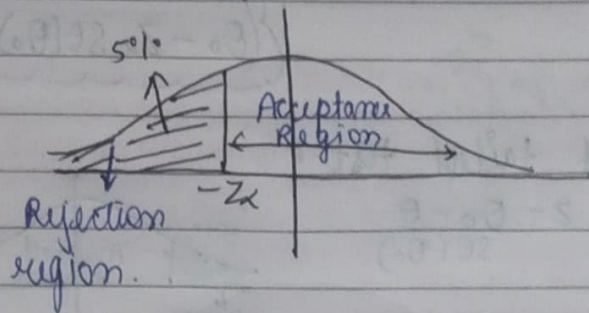
$\theta_0 \neq \theta$  (5%)  
 $\theta_0 < \theta$  (2.5%)  
 $\theta_0 > \theta$  (2.5%)



#  $H_1 : \theta_0 < \theta$  left tailed test

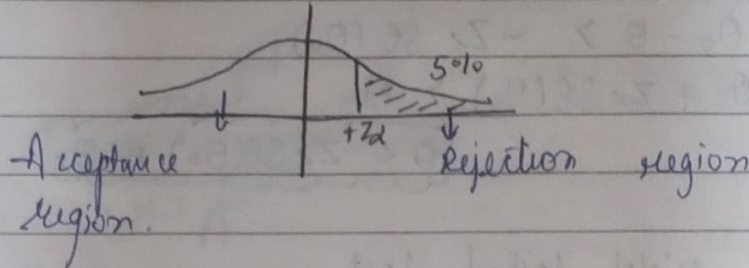
$$z = \frac{\theta_0 - E(\theta)}{SE(\theta_0)} \quad (\text{negative})$$

5% area on the left side only.



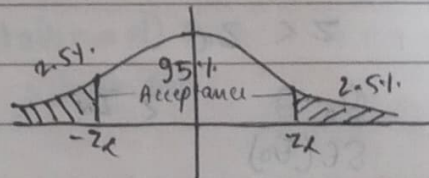
#  $H_1 : \theta_0 > \theta$  Right tailed test.

5% area on right side only.



## # Confidence Interval

# For two tailed test :



$$P(|Z| < z_\alpha) = 0.95$$

$$P\left(\left|\frac{\theta_0 - \theta}{SE(\theta_0)}\right| < z_\alpha\right) = 0.95$$

$$|\theta_0 - \theta| < z_\alpha SE(\theta_0)$$

$$-z_\alpha SE(\theta_0) < \theta_0 - \theta < z_\alpha SE(\theta_0)$$

|   | A   | B    | C    |
|---|-----|------|------|
| A | 0.4 | 0.15 | 0.15 |
| B | 0.1 | 0.8  | 0.1  |
| C | 0.2 | 0.2  | 0.6  |

$$-\theta_0 - Z_{\alpha} SE(\theta_0) < -\theta < -\theta + Z_{\alpha} SE(\theta_0)$$

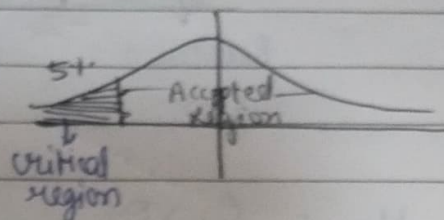
$$\theta_0 - Z_{\alpha} SE(\theta_0) < \theta < \theta_0 + Z_{\alpha} SE(\theta_0)$$

Therefore the confidence interval is:

$$(\theta_0 - Z_{\alpha} SE(\theta_0), \theta_0 + Z_{\alpha} SE(\theta_0))$$

(ii) For ~~left~~ right tailed test:

$$z = \frac{\theta_0 - \theta}{SE(\theta_0)}$$



$$z > -Z_{\alpha}$$

$$\frac{\theta_0 - \theta}{SE(\theta_0)} > -Z_{\alpha}$$

$$\theta_0 - \theta > -Z_{\alpha} SE(\theta_0)$$

$$\theta_0 + Z_{\alpha} SE(\theta_0) > \theta$$

$$\theta < Z_{\alpha} SE(\theta_0) + \theta_0$$

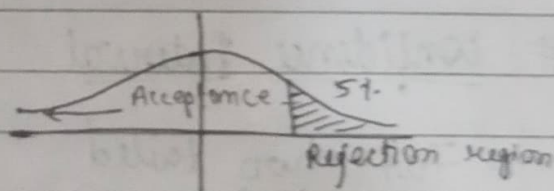
(iii) For right tailed test

$$z < Z_{\alpha}$$

$$\frac{\theta_0 - \theta}{SE(\theta_0)} < Z_{\alpha}$$

$$\theta_0 - \theta < Z_{\alpha} SE(\theta_0)$$

$$\theta_0 - Z_{\alpha} SE(\theta_0) < \theta$$



Confidence interval shows the range of parameter

—\* continued at start\*—

## ★ Critical values

| Type of Test      | LOS $\alpha$          |                       |                       |
|-------------------|-----------------------|-----------------------|-----------------------|
|                   | 1%                    | 2%                    | 5%                    |
| Two tailed Test   | $ Z_{\alpha}  = 2.58$ | $ Z_{\alpha}  = 2.33$ | $ Z_{\alpha}  = 1.96$ |
| Right tailed Test | $Z_{\alpha} = 2.33$   | $Z_{\alpha} = 2.055$  | $Z_{\alpha} = 1.645$  |
| Left tailed Test  | $Z_{\alpha} = -2.33$  | $Z_{\alpha} = -2.055$ | $Z_{\alpha} = -1.645$ |

# Large Sample Theory:

Process of testing of hypothesis:

Step 1: setup null and alternative hypothesis

$$H_0: \theta_0 = \theta \quad \text{no significant difference b/w } \theta_0 \text{ and } \theta$$

$$H_1: \begin{cases} \theta_0 \neq \theta \\ \theta_0 < \theta \\ \theta_0 > \theta \end{cases} \quad \left. \begin{array}{l} \text{any one of them} \\ \text{significant difference b/w } \theta_0 \text{ and } \theta \end{array} \right\}$$

Step 2: Set level of significance and find the critical value  $Z_{\alpha}$  (if not given then 5%).Step 3: Calculate the test statistic

$$Z = \frac{\theta_0 - E(\theta_0)}{SE(\theta_0)}$$

Step 4: Conclusion

→ If  $|Z| < |Z_{\alpha}|$  then  $H_0$  is accepted  
 $H_1$  rejected.

→ If  $|Z| > |Z_{\alpha}|$  then  $H_1$  is accepted  
 and  $H_0$  is rejected.

Test I Test of the significance of the difference b/w the population proportion and sample proportion.

Let  $P$  be the population proportion and  $p$  be the sample proportion

The test statistic is  $Z = \frac{p - E(p)}{SE(p)} = \frac{p - P}{SE(p)}$

$$= \frac{p - P}{\sqrt{\frac{PQ}{n}}} \quad Q = 1 - P$$

$n = \text{sample size}$   
 $n > 30$

Proof

$$\text{mean} = nP$$

$$\text{var} = nPQ$$

$$\text{var}(aX) = a^2 \text{Var}(X)$$

$$p = \frac{X}{n}$$

$$E(p) = \frac{1}{n} E(X) = \frac{1}{n} nP$$

$$\text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2} \text{var}(X) = \frac{1}{n^2} nPQ = \frac{PQ}{n}$$

Ex-8.1 1.

Sol<sup>n</sup>:

$$P \geq 0.90 \text{ (practically)}$$

$$n = 400$$

$$p = \frac{320}{400} = 0.8$$

$$H_0: p = P \geq 0.90$$

i.e. atleast 90% of voters support the candidate.

$$H_1: p < P \text{ i.e. less than 90\% of voters support the candidate. (left tailed).}$$

$$\text{LOS } \alpha = 0.05$$

$$Z_{\alpha} = -1.645 \text{ (check table).}$$

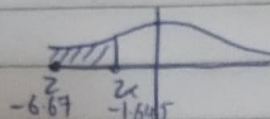
The test statistic

$$Z = \frac{p - P}{SE(p)} = \frac{0.8 - 0.9}{\sqrt{\frac{0.9 \times 0.1}{400}}} = -6.67$$

Since  $Z = -6.67$  is less than  $Z_{\alpha} = -1.645$

$$Z = -6.67 < Z_{\alpha} = -1.645$$

$H_0$  is rejected and  $H_1$  is accepted.



95% confidence interval.

$$Z \geq -1.645$$

$$\frac{p - P}{SE(p)} \geq -1.645$$

$$\frac{0.8 - P}{\sqrt{\frac{P(1-P)}{n}}} \geq -1.645$$

(answer may not come).  
So  $P(0 \rightarrow p_0)$

$$\frac{0.8 - P}{\sqrt{\frac{0.8 \times 0.2}{400}}} \geq -1.645$$

$$\frac{0.8 - P}{0.02} \geq -1.645$$

$$0.8 - P \geq -0.03$$

$$0.8 + 0.03 \geq P$$

$$0.83 \geq P$$

$$[P \leq 0.83]$$

8.1 3 Experience ---- quality.

$$P = 0.20 \quad n = 400 \quad p = \frac{50}{200} = 0.13$$

$H_0 \Rightarrow p = P = 0.20$  i.e. percentage of top quality is 20%.

$H_1 \Rightarrow p \neq P = 0.20$  i.e. 20% are not top quality

Two tailed

$$\text{LOS } \alpha = 0.01 \quad |Z_{\alpha}| = 2.58.$$

$$\text{Test statistic: } Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.13 - 0.20}{\sqrt{\frac{0.2 \times 0.8}{400}}} = -3.50$$

Since  $|Z| = 3.5 > |Z_{\alpha}| = 2.58$

$H_0$  is rejected  $H_1$  is accepted.  
i.e. the claim of 20% is wrong.

99% confidence interval for percentage of top quality.

$$\left| \frac{p - P}{SE(p)} \right| \leq |Z_{\alpha/2}|$$

Ans  
Confidence interval  
is  $[0.09, 0.17]$

$$\left| \frac{0.13 - P}{\sqrt{\frac{pq}{n}}} \right| \leq 2.58$$

$$|0.13 - P| \leq 2.58 \sqrt{\frac{0.13 \times 0.87}{400}}$$

$$|0.13 - P| \leq 0.04$$

$$-0.04 \leq 0.13 - P \leq 0.04$$

$$-0.17 \leq -P \leq -0.09 \Rightarrow 0.09 \leq P \leq 0.17$$

Test: n

Test of significance of difference between two sample proportion

 $n_1$  $p_1$  $n_2$  $p_2$  $P$ 

{ sample population }

$$E(p_1 - p_2) = E(p_1) - E(p_2)$$

$$= P - P$$

$$= 0$$

 $\textcircled{P_1}$  $\textcircled{P_2}$  $\downarrow$  $\downarrow$  $p_1$  $p_2$ 

{ different population }

$$= P_1 - P_2 \neq 0 -$$

★

$$Z = \frac{p_1 - p_2 - E(p_1 - p_2)}{SE(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

Ex 8.2 Ques 1

(a)

$$n_1 = 150$$

$$p_1 = \frac{63}{150} = 0.42$$

$$n_2 = 200$$

$$p_2 = \frac{60}{200} = 0.30$$

$H_0 : p_1 = p_2$  there is no significance difference between recall prop. of the commercials

$H_1 : p_1 \neq p_2$  i.e. there is significant diff. between recall proportion of two commercials (two tailed)

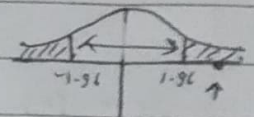
$$\alpha = 0.05 \quad |Z_{\alpha}| = 1.96$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{63 + 60}{350} = 0.35$$

$$Z = \frac{p_1 - p_2}{\sqrt{pq \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.42 - 0.30}{\sqrt{0.35 \times 0.65 \left( \frac{1}{150} + \frac{1}{200} \right)}}$$

$$= 2.33$$

Since  $|Z| = 2.33 > |Z_{0.05}| = 1.96$ .  
 $H_0$  is rejected  $H_1$  is accepted.



The 95% confidence interval for the difference between population proportion is  $P_1 - P_2$  is.

$$(p_1 - p_2 - SE(p_1 - p_2)1.96, p_1 - p_2 + SE(p_1 - p_2)1.96)$$

$$\Rightarrow (0.12 - 0.01, 0.12 + 0.01)$$

$$\Rightarrow [0.11, 0.13]$$

This implies  $P_1 - P_2 \in (0.11, 0.13)$

Ques 2 A machine - - - -

$$n_1 = 400$$

$$p_1 = 20/400 = 0.05$$

$$n_2 = 300$$

$$p_2 = 10/300 = 0.03$$

$$H_0: p_1 = p_2$$

The machine has not improved.

$$H_1: p_1 > p_2$$

machine has improved.

$$\alpha = 0.05$$

$$Z_{0.05} = 1.645$$

$$P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = 0.04$$

$$p =$$

$$Z = \frac{p_1 - p_2}{\sqrt{pq \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.05 - 0.03}{\sqrt{0.04 \times 0.96 \left( \frac{1}{400} + \frac{1}{300} \right)}}$$

$$Z = 1.34$$

Machine has not improved.

$$\text{Since } Z = 1.34 < Z_{\alpha} = 1.645$$

$H_0$  accepted  
 $H_1$  rejected.

Test 3

Test of significance difference between sample mean and population mean.

$\bar{X}$  → sample mean

$n$  → sample size.

$$Z = \frac{\bar{X} - \mu}{SE(\bar{X})} \begin{cases} \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} & \text{if } \sigma \text{ is known} \\ \frac{\bar{X} - \mu}{s/\sqrt{n}} & \text{if } \sigma \text{ is not known} \end{cases}$$

$\mu$  → pop. mean.

$\sigma$  → pop. s.d.

$s$  → sample s.d.

8.3

quest  $\mu = 260$        $n = 150$        $\bar{X} = 260 + 20 = 280$

$H_0$  :  $\bar{X} = \mu = 260$       The display did not help much.

$H_1$  :  $\bar{X} > \mu$       difference is by chance  
the display did help much.

$\alpha = 0.05$        $Z_{0.05} = 1.645$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{280 - 260}{45/\sqrt{150}}$$

$$Z_{\text{calc}} = 5.44$$

$$Z = 5.44 > Z_{\alpha} = 1.645$$

$H_0$  is rejected and  $H_1$  is accepted.

display did help much.

# Test IV Test of significance of the difference between two sample means.

Two samples :

|       |             |       |
|-------|-------------|-------|
| $n_1$ | $\bar{X}_1$ | $S_1$ |
| $n_2$ | $\bar{X}_2$ | $S_2$ |

drawn from a population with mean  $\mu$  and SD  $\sigma$  or from two different populations with the means  $\mu_1$  and  $\mu_2$  and SD  $\sigma_1$  and  $\sigma_2$  respectively

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad \text{if } \sigma_1 \neq \sigma_2 \text{ known.}$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{if } \sigma_1 = \sigma_2 = \sigma \text{ known}$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \quad \text{if } \sigma_1 \text{ and } \sigma_2 \text{ not known } \sigma_1 \neq \sigma_2$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \quad \text{if } \sigma_1 = \sigma_2 = \sigma \text{ not known}$$

$$\sigma^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}$$

Ex. 8.4 ans  $n_1 = 56$   $\bar{X}_1 = 24.25$   $S_1 = 10.2$   
 $n_2 = 50$   $\bar{X}_2 = 20.25$   $S_2 = 9.7$

$H_0$   $\bar{X}_1 = \bar{X}_2$  i.e. program is equally effective

$H_1$   $\bar{X}_1 > \bar{X}_2$  i.e. program more effective for male

$$\alpha = 0.02 \quad Z_{0.02} = 2.055$$

since the samples are from same population

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{24.25 - 20.25}{\sqrt{\frac{(10.2)^2}{50} + \frac{(9.7)^2}{56}}} = 2.06$$

since  $(Z = 2.06) = (Z_{\alpha} = 2.055)$  then  $H_0$  is accepted and  $H_1$  is rejected.

Ques 2

$$n_1 = 250$$

$$\bar{X}_1 = 120$$

$$S_1 = 14$$

$$n_2 = 400$$

$$\bar{X}_2 = 124$$

$$S_2 = 14$$

$$H_0: \bar{X}_1 = \bar{X}_2$$

ie. no significant diff b/w avg weight

$$H_1: \bar{X}_1 \neq \bar{X}_2$$

ie. significant diff b/w avg weight.

$$\alpha = 0.01$$

$$|Z_{0.01}| = 2.58$$

since the sample are from two different populations with unknown SD.

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{120 - 124}{\sqrt{\frac{14^2}{250} + \frac{14^2}{400}}} = -3.87$$

$$\text{since } |Z| = 3.87 > |Z_{\alpha}| = 2.58$$

$H_0$  is rejected and  $H_1$  is accepted

the difference is significant.

The confidence interval for the difference  $\mu_1 - \mu_2$   
 $((\bar{X}_1 - \bar{X}_2) - SE(\bar{X}_1 - \bar{X}_2)2.58, (\bar{X}_1 - \bar{X}_2) + SE(\bar{X}_1 - \bar{X}_2)2.58)$   
 $(-4 - 2.66, -4 + 2.66) \Rightarrow (-6.66, -1.34) \text{ for } (\mu_1 - \mu_2)$   
 $(-1.44, 6.66)$

Test IV Test of significance of the difference b/w the population sd  $\sigma$  and sample sd  $S$ .

$$Z = \frac{S - \sigma}{\sigma / \sqrt{n}}$$

Test V Test of significance diff. between two samples

$$Z = \frac{S_1 - S_2}{\sigma \sqrt{\frac{1}{2n_1} + \frac{1}{2n_2}}}$$

$$Z = \left\{ \begin{array}{l} \frac{S_1 - S_2}{\sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}} \quad \text{if } \sigma_1 \neq \sigma_2 \text{ if known.} \end{array} \right.$$

$$\frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_1} + \frac{S_2^2}{2n_2}}} \quad \text{if } \sigma_1 \neq \sigma_2 \text{ not known. } S_1, S_2 \text{ known.}$$

$$\sigma \sqrt{\frac{1}{2n_1} + \frac{1}{2n_2}} \quad \text{if } \sigma_1 = \sigma_2 \text{ known.}$$

$$\frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_1} + \frac{S_2^2}{2n_2}}} \quad \text{if } \sigma_1 = \sigma_2 = \sigma \text{ not known.}$$

$$\sigma^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}$$

Q. 8.5. Ques 2

-----  $\sigma = 100h$   $n = 150$   $S = 95h$

$\alpha = 0.05$

$H_0$   $S = \sigma$  i.e. manufacturing not justified by changing process.

$H_1$   $S < \sigma$  i.e. manufacturing is justified by changing process.

$$\alpha = 0.05$$

$$Z_{0.05} = -1.645$$

$$Z = \frac{95 - 100}{\frac{100}{\sqrt{300}}} = -0.87$$

Since  $Z = -0.87 > Z_{\alpha} = -1.645$  this implies  $H_0$  is accepted and  $H_1$  is rejected.

ans 4

$$n_1 = 1000$$

$$S_1 = 2.6$$

$$n_2 = 500$$

$$S_2 = 2.7$$

$$H_0: S_1 = S_2 = \sigma$$

$$H_1: S_1 \neq S_2 \quad (\text{i.e. } \sigma_1 \neq \sigma_2)$$

$$\alpha = 5\%$$

$$|Z_{\alpha}| = 1.96$$

$$Z = \frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_1} + \frac{S_2^2}{2n_2}}} = \frac{2.6 - 2.7}{\sqrt{\frac{(2.6)^2}{1000} + \frac{(2.7)^2}{2000}}} = -0.98$$

# Small Sample Test

sample size  $n < 30$ .

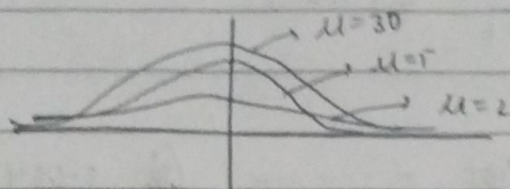
t-distribution

A random variable  $t$  is said to follow  $T$ -distribution if its probability density function is given by

$$f(t) = \frac{1}{\sqrt{\pi} \beta \left( \frac{\mu}{2}, \frac{1}{2} \right)} \left( 1 + \frac{t^2}{\mu} \right)^{-(\mu+1)/2}$$

$-\infty < t < \infty$

where  $\mu$  denotes degree of freedom of  $T$  for  $\mu > 30$   $T$  distribution becomes normal distribution.  $\mu \downarrow \rightarrow$  flatten.



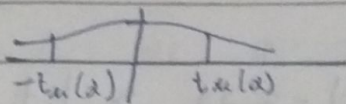
$$x_1 + x_2 + x_3 = 50$$

$$\hookrightarrow \mu = 3 - 1 = 2$$

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \rightarrow n \rightarrow \infty$$

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 \quad \gamma = n-1$$

$\alpha$  - level of significance,  $\mu$  -  $t_{\mu}(\alpha)$



Two tailed

$$|t_{\mu}(0.04)| = 2.3281$$

One tailed

$$\text{right } t_{\mu}(0.02) = 2.3281$$

$$\text{left } t_{\mu}(0.02) = -2.3281$$

★ Test I : Test of significance difference of population mean and sample mean.

Let  $\bar{x}$  be the mean of a small sample of size  $n$  and stand. deviation  $s$ . drawn from a normal population with mean  $\mu$  then :

$$t = \frac{\bar{x} - E(\bar{x})}{SE(\bar{x})} = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} \quad \text{at } \mu = n-1$$

where

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \quad \left\{ \begin{array}{l} \text{from theory of} \\ \text{estimation} \end{array} \right\}$$

$$E(S^2) = \frac{1}{n-1} E(n\sigma^2)$$

Ex 8.6

Quest 1 A pharmaceutical ----- ( $t_{\alpha} = 2.064$ ).

$$\bar{x} = 7 \text{ mmHg}$$

$$s = 1.5 \text{ mmHg}$$

$$\mu = 8 \text{ mmHg}$$

$$n = 25$$

$$H_0 : \bar{x} = \mu$$

$$H_1 : \bar{x} < \mu$$

the drugs effectiveness is as claimed

----- x ----- is not as claimed

$$\alpha = 0.05$$

$$N = n-1 = 24$$

$$t_{24}(0.05) = 1.7109$$

Left  
Tailed  
Test

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{7 - 8}{\frac{1.5}{\sqrt{24}}} = -3.27$$

$$|t| = 3.27 > |t_{24}(0.05)| = 1.7109$$

$H_0$  is rejected and  $H_1$  is ~~not~~ accepted.

The 95% Interval for  $\mu$  is

$$\frac{\bar{x} - \mu}{s/\sqrt{n-1}} > -1.7109$$

$$\frac{7 - \mu}{\frac{1.5}{\sqrt{24}}} > -1.7109$$

$$\frac{\mu - 7}{1.5/\sqrt{24}} < 1.7109$$

$$\boxed{\mu < 7.52}$$

Ques 2Soln.

$$n = 8, \quad \bar{X} = \frac{\quad}{8} = 12.10$$

$$N = 8 - 1 = 7$$

$$H_0: \bar{X} = \mu = 11.70$$

$$H_1: \bar{X} \neq \mu$$

$$|t_7(0.02)| = 2.9980$$

$$S^2 = \frac{1}{n-1} \sum (x_i - \bar{X})^2 = \frac{(12.2 - 12.1)^2 + \dots + (12.4 - 12.1)^2}{7}$$

$$S^2 = 0.11$$

$$t = \frac{\bar{X} - \mu}{\sqrt{\frac{S^2}{n-1}}} = \frac{12.10 - 11.7}{\sqrt{\frac{0.11}{7}}} = 3.19$$

$H_0$  rejected and  $H_1$  accepted.  
The mean weight cannot be 11.7 grams.

98% confidence interval for  $\mu$  is.

$$\Rightarrow \left( \bar{X} - SE(\bar{X}) t_7(0.02), \bar{X} + SE(\bar{X}) t_7(0.02) \right)$$

$$\Rightarrow \left( 12.10 - \sqrt{\frac{0.11}{7}} \times 2.9980, 12.10 + \sqrt{\frac{0.11}{7}} \times 2.9980 \right)$$

$$\Rightarrow (11.72, 12.48)$$

Test II : Test of significance of difference of two sample means (Independent samples).

$$\begin{array}{ccc} n_1 & \bar{X}_1 & S_1 \\ n_2 & \bar{X}_2 & S_2 \end{array}$$

$$* \quad t = \begin{cases} \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2} \left[ \frac{1}{n_1} + \frac{1}{n_2} \right]}} & \text{if } n_1 \neq n_2 \\ & V = n_1 + n_2 - 2 \\ \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2 + S_2^2}{n - 1}}} & \text{if } n_1 = n_2 = n \\ & V = 2n - 2. \end{cases}$$

Ex. 7 Ques 3

$$n_1 = 12 \quad n_2 = 14$$

$$\bar{A} = 5.75$$

$$\bar{B} = 5$$

$H_0: \bar{X}_1 = \bar{X}_2 \quad \bar{A} = \bar{B}$  Both vaccines have same side effects.

$$H_1: \bar{A} > \bar{B}$$

$$S_1^2 = \frac{1}{n_1 - 1} \sum (A_i - \bar{A})^2 = 4.75$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (B_i - \bar{B})^2 = 4.76, 4.77.$$

$$\alpha = 0.01$$

$$V = 12 + 14 - 2 = 24$$

$$t_{24}(0.01) = 2.4922$$

$$t = \frac{\bar{A} - \bar{B}}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left[ \frac{1}{n_1} + \frac{1}{n_2} \right]}}$$

$$= \frac{5.75 - 5}{\sqrt{\frac{12 \times 4.75 + 14 \times 4.77}{12 + 14 - 2} \left[ \frac{1}{12} + \frac{1}{14} \right]}}$$

$$= 0.84$$

Since  $t = 0.84 < t_{24}(0.01) = 2.4922$   
 $H_0$  is accepted.

Two vaccines have same side effects.

Test: III Test of significance of difference between the means of two dependent samples.

Then

$$d_i = x_i - y_i$$

$$\bar{d} = \frac{\sum d_i}{n} = \bar{x} - \bar{y}$$

$$s^2 = \frac{1}{n-1} \sum (d_i - \bar{d})^2$$

The test statistic is

$$* t = \frac{\bar{d}}{s/\sqrt{n-1}}$$

Ex 8-8 Quest Ten soldiers - - - - -

$$n = 10$$

|                   |                    |       |       |      |        |       |       |      |       |       |
|-------------------|--------------------|-------|-------|------|--------|-------|-------|------|-------|-------|
| X                 | $\Rightarrow$ 67   | 24    | 57    | 55   | 63     | 54    | 56    | 68   | 33    | 43    |
| Y                 | $\Rightarrow$ 70   | 38    | 58    | 58   | 56     | 67    | 68    | 72   | 42    | 38    |
| $d_i = x_i - y_i$ | $\Rightarrow$ -3   | -14   | -1    | -3   | 7      | -13   | -12   | -4   | -9    | 5     |
| $(d_i + 4.7)^2$   | $\Rightarrow$ 2.89 | 86.49 | 13.69 | 2.89 | 136.89 | 68.89 | 53.29 | 0.49 | 18.49 | 94.09 |

$$\sum d_i = -47$$

$$\bar{d} = -47/10 = -4.7$$

$$\sum (d_i - \bar{d})^2 = 478.91$$

$$s^2 = \frac{1}{n-1} \sum (d_i - \bar{d})^2 = \frac{478.91}{9} = 53.12$$

$$H_0 : \bar{d} = 0 \quad (\text{i.e. } \bar{x} = \bar{y})$$

$$H_1 : \bar{d} < 0 \quad (\bar{x} < \bar{y})$$

$$\alpha = 0.05$$

$$\nu = n-1 = 9$$

$$t_{\alpha}(0.05) = 1.8331$$

$$t = \frac{\bar{d}}{\sqrt{s^2/n-1}} = \frac{\bar{d}}{\sqrt{\frac{s^2}{n-1}}} = \frac{-4.7}{\sqrt{\frac{53.12}{9}}} = -1.93$$

$$\text{Since } |t| = 1.93 > |t_{\alpha}(0.05)| = 1.8331$$

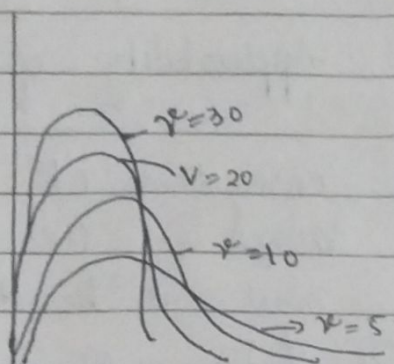
$\Rightarrow H_0$  is rejected,  $H_1$  accepted.

## $\chi^2$ - test (chi-square test)

The square of a normal variable  $x$  is called  $\chi^2$ -variate,  $\chi^2$  is said to follow the  $\chi^2$  distribution with the degrees of freedom  $\nu$  if its pdf is

$$f(\chi^2) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} (\chi^2)^{\nu/2 - 1} e^{-\chi^2/2}$$

$$0 < \chi^2 < \infty$$



### Test I $\chi^2$ as a test of goodness of fit.

If  $x_1, x_2, x_3, \dots, x_n$  are the normal variable then  $\chi^2 = x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2$  follows approximately the  $\chi^2$  distribution with degrees of freedom  $\nu = n$ .

The test of goodness of fit measures how good observed frequencies fit the expected frequencies.

Let  $O_1, O_2, O_3, \dots, O_n$  and  $E_1, E_2, E_3, \dots, E_n$  be the observed and its expected frequencies in some experiment, such that.

$$\sum_{i=1}^n O_i = \sum_{i=1}^n E_i \quad \nu = n-1$$

The best statistic is :

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad \text{at } \nu = n-1$$

under the hypothesis.

$$H_0 = O_i = E_i$$

$$H_1 = O_i \neq E_i$$

Condition for Applicability of  $\chi^2$

- (i) The number of observations  $N$  in sample must be reasonably large (i.e.  $N > 10$ )
- (ii) Individual freq. must not be too small.  
 $O_i \neq 10$  or  $O_i \geq 10$
- (iii) The number of classes  $n$  must be neither too small nor too large i.e.  $4 \leq n \leq 16$ .

Ex 8.9

Ques 2 The demand ---- week.

Sol<sup>n</sup>. Let  $H_0 \Rightarrow O_i = E_i$  demand does not depend on day.

$H_1 \Rightarrow O_i \neq E_i$  demand depends on day

If the demand does not depend on days then each day the demand should be

$$= \frac{\text{---}}{6} = 1120$$

$E_i = 1120$  for all days.

|       |      |      |      |      |      |      | Total |
|-------|------|------|------|------|------|------|-------|
| $O_i$ | 1124 | 1125 | 1110 | 1120 | 1126 | 1115 | 6720  |
| $E_i$ | 1120 | 1120 | 1120 | 1120 | 1120 | 1120 | 6720  |

$$\alpha = 0.05 \quad \nu = n-1 = 5$$

$$\chi^2_{(0.05)} = 11.07$$

$$\chi^2 = \sum_{i=1}^6 \left[ \frac{(O_i - E_i)^2}{E_i} \right]$$

$$= \frac{1}{1120} [16 + 25 + 100 + 36 + 25]$$

$$\chi^2 = 0.1875$$

Since  $\chi^2 = 0.18 < \chi^2_{(0.05)} = 11.07$   
 $\Rightarrow$   $H_0$  is accepted.  
 The demand does not depends on days.

Ques 5 Genetic - - - - 5%

| Type  | M  | NN  | N  | Total |
|-------|----|-----|----|-------|
| $O_i$ | 90 | 135 | 75 | = 300 |
| $E_i$ | 75 | 150 | 75 | = 300 |
|       | 1  | 2   | 1  |       |

$H_0$   $O_i = E_i$

$H_1$   $O_i \neq E_i$

the sample confirms theory.  
 the sample does not confirm theory.

$$\alpha = 0.05$$

$$\nu = n-1 = 2$$

$$\chi^2_{(0.05)} = 5.991$$

$$\chi^2 = \frac{1}{300} \sum_{i=1}^3 \frac{(O_i - E_i)^2}{E_i} = \frac{225}{75} + \frac{225}{150} = 4.5$$

$$\chi^2 = 4.5 < \chi^2_{(0.05)} = 5.991$$

$H_0$  is accepted.

$\chi^2$  - test on independence of attributes:

Let the total No.  $N$  of observations is divided into  $m \times n$  classes according to some attributes of classification.

Contingency  
Table

|             |          | Attribute B |          |          |       |          | Total |
|-------------|----------|-------------|----------|----------|-------|----------|-------|
|             |          | $B_1$       | $B_2$    | $B_3$    | ----- | $B_n$    |       |
| Attribute A | $A_1$    | $O_{11}$    | $O_{12}$ | $O_{13}$ | ----- | $O_{1n}$ | $R_1$ |
|             | $A_2$    | $O_{21}$    | $O_{22}$ | $O_{23}$ | ----- | $O_{2n}$ | $R_2$ |
|             | $A_i$    | $O_{31}$    | $O_{32}$ | $O_{33}$ | ----- | $O_{3n}$ | $R_3$ |
|             | $\vdots$ | $\vdots$    | $\vdots$ | $\vdots$ |       | $\vdots$ |       |
|             | $A_m$    | $O_{m1}$    | $O_{m2}$ | $O_{m3}$ |       | $O_{mn}$ | $R_m$ |
| Total       |          | $C_1$       | $C_2$    | $C_3$    | ----- | $C_n$    | $N$   |

Then:

$$P(A_i) = \text{Probability of having the attributes } A_i \text{ only}$$

$$* = \frac{R_i}{n}$$

$$P(B_j) = \text{Probability of having the attributes } B_j \text{ only}$$

$$* P(B_j) = \frac{C_j}{n}$$

$P(A_i B_j)$  : The probability of having both  $A_i$  and  $B_j$  attributes

$$= P(A_i) P(B_j) \quad \text{if } A_i \text{ and } B_j \text{ are independent.}$$

$$* P(A_i B_j) = \frac{R_i \times C_j}{N^2}$$

Therefore the expected no. of observations which are independent of  $A_i$  and  $B_j$

$$* E_{ij} = N \cdot \frac{R_i C_j}{N^2} = \frac{R_i C_j}{N}$$

Therefore under the hypothesis

$$H_0 = O_{ij} = E_{ij} \quad \text{i.e. the attributes are independent.}$$

$$H_1 = O_{ij} \neq E_{ij} \quad \text{i.e. the attributes are not independent.}$$

The test statistic is :

$$\chi^2 = \sum_{i=1}^m \sum_{j=1}^n \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad \text{at } v = \frac{(m-1)(n-1)}{1}$$

Ex 8.16

Ques 2Sol<sup>n</sup>

$$H_0 \Rightarrow O_{ij} = E_{ij}$$

the performance does not depend on family structure.

$$H_1 \Rightarrow O_{ij} \neq E_{ij}$$

the performance depends on Family structure.

The expected no. of students in each table:

$$E_{ij} = \frac{R_i \times C_j}{N}$$

The contingency table is:

Academic Status

|            | Academic Status |            |            |       |
|------------|-----------------|------------|------------|-------|
|            | School          | Graduate   | Post grad. | Total |
| No Parent  | 15 (34.44)      | 32 (23.56) | 35 (24.00) | 82    |
| One Parent | 92 (80.65)      | 52 (55.15) | 48 (56.20) | 192   |
| Two Parent | 48 (39.91)      | 22 (27.29) | 25 (27.80) | 95    |
| Total      | 155             | 106        | 108        | 369   |

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{82 \times 155}{369} = 34.44$$

$$E_{22} = \frac{106 \times 192}{369}$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{155 \times 192}{369} = 80.65$$

$$E_{23} = \frac{106 \times 95}{369}$$

$$E_{31} = \frac{155 \times 95}{369} = 39.91$$

$$E_{21} = \frac{106 \times 82}{369} = 23.56$$

then

$$\chi^2 = \sum_{i=1}^3 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$= 10.97 + 1.60 + 1.64 + 3.02 + 0.18 + 1.03 \\ + 5.04 + 1.20 + 0.28$$

$$\chi^2 = 24.96$$

$$N = 2 \times 2 = 4$$

$$\alpha = 0.05$$

$$\chi^2_{4(0.05)} = 9.49$$

since

$$\chi^2 = 24.96 > \chi^2_{4(0.05)} 9.49.$$

$H_0$  is rejected  $H_1$  is accepted.

T-Test

IV

Test of Independence of population and sample standard deviations.

$$N < 30$$

S

$$t = \frac{s - \bar{s}}{s/\sqrt{n}}$$